

Course Presentation

Advanced Programming – Programación Avanzada

GIT (Ob 380001)

Dra. M^a Dolores Rodríguez Moreno

Teachers and Tutoring hours

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Tutoring hours: Mon: 10-12

Wed: 16-18

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Schedule (GIT)

- Monday: 12-14 (Theory) NORTE A2
 - Wednesday: 8 a 10, 10 a 12 & 12 a 14 (lab)
 - LAB: E L6
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- First 2 weeks (NO LABS)
 - Theory: ESTE A6
 - Time: Wednesday 10-12

General features

Degree	GIT
Code	380001
Course	3rd
	Compulsory
Semester	First
Credits	6 ECTS
Web	Aula Virtual http://uah.blackboard.com
	All documentation and dates available

Objectives

General Objective

Learn the logic of object-oriented programming (OOP)

Specific Objectives

- Classes and Objects
- Operator Overloading
- Inheritance
- Polymorphism
- File I/O (Input/Output) Streams
- Using and Creating Object Libraries (Modules and Packages)



Contents

Part	Topics
1. OOP 2. Classes	• Principles and Logics • Constructors, Methods, attributes, access control • Lists, tuples, sets, dictionaries... • Implementation of special mths
3. Data structures in Python	
4. Derived Classes	
5. Exceptions 6. File (I/O)	• Single and multiple inheritance • Error handling • Reading /writing files • Database

Course Scheduling

- Time distribution of the course:



Methodology

- Theory classes
- Problems solved
- Flipped Learning supported by LLMs



Lab Practice Regulations

- The **submission schedule** will be specified for each practice in the course schedule
- The statement of each practice will be available on the Aula Virtual (uah.blackboard.com)
- Practices will be carried out according to the planned schedule; this means that holidays/exams will not alter the calendar
- The student does not need to submit the laboratory exercises (PL) on the date established in the calendar
- The grade for the PL will be exclusively the final exam score

Continuous Assessment

Instruments Rating	% Mark
1. Final Exam - PEF	40%
2. Testing Lab - PRL	30%
3. Partial Exam - PEI	30%

Testing Lab (30%)

- Each lab assignment has a deadline:
- You can do it in pairs
- The **lab exam** defines the **lab mark**
- The evaluation of each practical case includes the practice to be carried out and the theory related to it
- There will be 1 exam of the lab → 30%
- NO PRESENTADO: any exam

Single Assessment: Final exam

- Consist of a single test on both theoretical and practical content, which will account for 100% of the course grade
- Students eligible to take this exam are those who have been granted the right to be evaluated through a final exam. The exam may be conducted either orally and/or in writing
- To pass the final exam, students must submit any practical work that has not yet been evaluated on the day of the final exam
- The evaluation of each practical case includes the practical work itself and the related theory

Extraordinary Assessment

- Students who have not passed the course in the regular exam session may opt for an extraordinary session, which will consist of a single test on both theoretical and practical content, accounting for 100% of the course grade
- This exam may be conducted either orally and/or in writing.
- To pass this final exam, students must submit their practical work before the day of the final exam
- The evaluation of each practical case includes both the practical work itself and the related theory.

Regulation

- Notifications by the e-learning platform
- Contact by e-mail
- Respect and courtesy
- Plagiarism/Copy in exams will not be tolerated

Bibliography

- Python Official Documentation:
<https://www.python.org/about/gettingstarted/>
- Object Oriented Programming with Python for Beginners: Mastering the Foundations of OOP. From Principles to Practice. 2 in 1 Guide.
Autor: SAM CAMPBELL
- Lenguajes de programación. Diseño e Implementación. Terence W.Pratt. Marvin V. Zelkowitz. Prentice Hall.

About me...

- During 2021-2023: Research Applied Center of The Netherlands (TNO)
- Full Professor (2018)
- European PhD in Computer Sciences (Awarded)
- Postdoc
 - NASA Ames, USA
 - JPL-NASA, USA
- Research interest: AI & ML, Robotics

TFG offer

- Interested in AI?
 - LLMs
 - Machine Learning
 - Robotics

Questions???

Dra. M^a Dolores Rodríguez Moreno